Class date: 14/12/2025
IWA

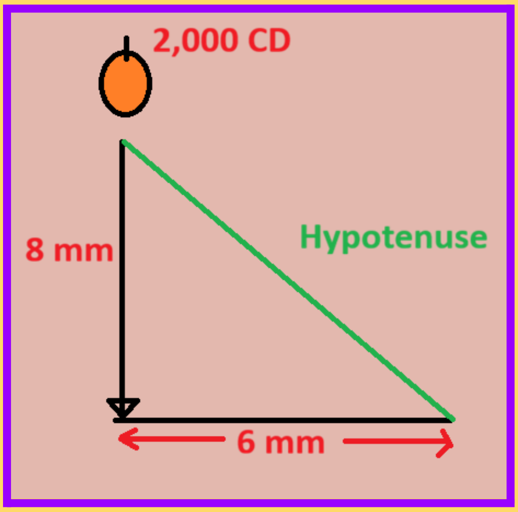
Lesson #6.00 2365 - Level 3 City and Guilds

Question example 3:

A street lighting column suspends a 2,000 cd light source. 8 metres above the ground. Determine the illuminance directly below the lamp and 6metres to one side of the lamp base.

Pre-requisites
Cd -> how bright the light is.
D^2 -> distance to the ground

Optional - Draw a basic picture



Calculate the luminance directly under lamp first

$$\text{Luminance} = \frac{\text{cd}}{d^2} = \frac{2000}{8^2} = \frac{2000}{64} = 31.25$$

= 31.35 lux

Class date: 14/12/2025
IWA

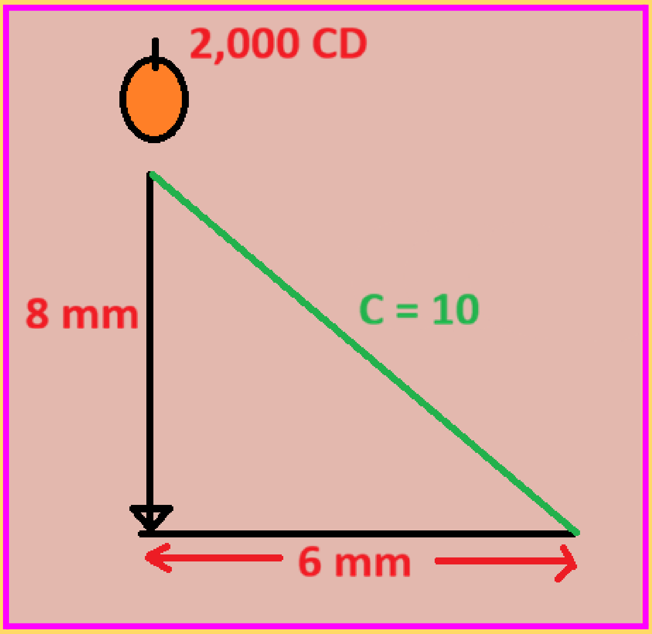
Then, Calculate the luminance on the angled line

First, Calculate the hypotenuse.

$$c = \sqrt{a^2 + b^2}$$

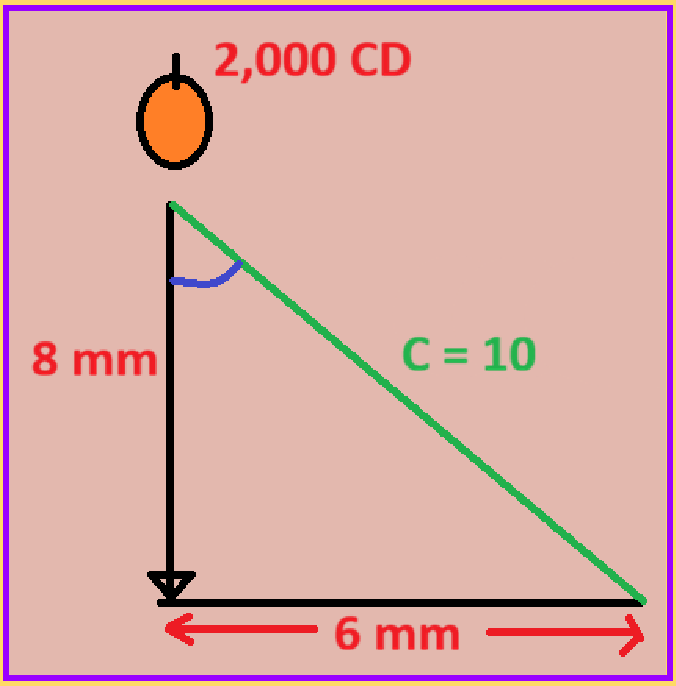
$$c = \sqrt{8^2 + 6^2}$$
$$c = \sqrt{64 + 36}$$
$$c = \sqrt{100}$$
$$c = 10$$

Update the picture with hypotenuse value



Class date: 14/12/2025
IWA

Next calculate the cosign.



The cosign is equal to the adjacent / hypotenuse.

Therefore, 8 / 10 => 0.8.

Finale

$$\boxed{\frac{\mathrm{cd}}{d^2} \times \cos\theta}$$

2,000 / 10^2 x 0.8 = **16 lux**